SIMATS Engineering • CSE • CSA0802 — Python Programming • 24-07-2026 • Category: Analytical • Student Submissions Report

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Q1. Harshad number

ANSWER

```
def is_harshad(n):
     s = sum(int(d) for d in str(n))
     return n % s ==0
n = int(input("enter a number:"))
if is_harshad(n):
   print(n,"is a harshad number")
else:
   print(n,"is not a harshad number")
print("harshad number between 1 and 500:")
for i in range(1,501):
     if is_harshad(i):
         print(i,end =" ")
```

Q2. GCD

CODE

```
a = int(input("enter first:"))
b = int(input("\nenter second:"))
while b≠0:
    temp =a
    a = b
    b = temp%b
print("\nGCD is",a)
```

INPUT

```
5
7
```

OUTPUT

```
enter first:
enter second:
GCD is 1
```

Q3. sum to a target value

ANSWER

```
seen = set()
for num in lst:
complement = target - num
if complement in seen:
seen.add(num)
return pairs
result = find_pairs(data,t)
```

Q4. Bitwise Operators

ANSWER

```
a = 13, b = 21, c = 26, result = 31
0b11111 0o37 0x1f
7 62 -32
a - int
b - int
c - int
result - int
```

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low = 0
    high = len(arr)-1
    while low ≤ high:
        mid = (low+high) // 2
        if arr[mid] == target:
            return mid
        if arr[low] ≤ arr[mid]:
            if target ≥ arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
                if target > arr[mid] and target ≤ arr[high]:
                    low = mid + 1
                else:
                    high = mid - 1
    return -1
data = [4,5,6,0,1,2]
t = int(input("target:"))
result = rotated_binary_search(data,t)
print("\nfound at:",result)
```

OUTPUT

```
target:
found at: 3
```